Eternal-Memory-Brain-Chip (EMBC) — Revised Mathematical Derivation and Physical Architecture

Idea-only, not tested or validated

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1 Scope, assumptions, notation, units

Scope: Provide a complete, self-consistent mathematical model (parametric) for computing layer composition and conditions that enable autonomous material feedback loops in the EMBC architecture. This document remains conceptual and **does not** include fabrication instructions.

Key assumptions (explicit):

$$\text{Linear, time-invariant, quasi-static electromagnetic approximations near the operating point.}$$

$$\text{BNNT are anisotropic; both isotropic EMT (MG/Bruggeman) and anisotropic EMT (tensorial) are provided.}$$

$$\text{Complex permittivity formalism used to capture losses; nonlinear and quantum effects excluded.}$$

Notation and units (selected):

$$v_{\mathrm{Si}},\ v_{\mathrm{BNNT}},\ v_{\mathrm{GaN}}$$

$$v_{\mathrm{Si}} + v_{\mathrm{BNNT}} + v_{\mathrm{GaN}} = 1$$

$$\varepsilon_0 = 8.854 \times 10^{-12}\ \mathrm{F/m}$$

$$\varepsilon_h,\ \varepsilon_{\mathrm{BNNT}\parallel},\ \varepsilon_{\mathrm{BNNT}\perp},\ \varepsilon_{\mathrm{GaN}},\ \varepsilon_{\mathrm{eff}}$$

$$\sigma_h,\ \sigma_{\mathrm{BNNT}},\ \sigma_{\mathrm{GaN}},\ \sigma_{\mathrm{eff}}\ (\mathrm{S/m})$$

$$L_{\mathrm{mat}}\ (\mathrm{H}),\quad C_{\mathrm{mat}}\ (\mathrm{F}),\quad f_0\ (\mathrm{Hz})$$

$$L_{\text{path}}\ (\text{m}), \quad v_p = \frac{c}{\sqrt{\varepsilon_{\text{eff}}}}, \quad t_p\ (\text{s})$$

$$\frac{|f - f_0|}{f_0} \le 1\%, \quad t_p \le 0.82\ \text{ns}$$

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2 Stoichiometry for YCore (symbolic + numeric example)

Project baseline formula:

$$\text{YCore} = n \cdot \text{C}_{12}\text{H}_{14}\text{O}_5\text{Si}(\text{F}_\text{x}) + 2 \cdot \text{XeN} + m \cdot (\text{SiR}_2\text{O})$$

Total molar mass:

$$M_{\text{YCore}}(n, m) = n\, M_{\text{org}} + 2\, M_{\text{XeN}} + m\, M_{\text{SiR2O}}$$

Mass-fraction constraints (two independent equations):

$$\frac{n\, M_{\text{org}}}{M_{\text{YCore}}} = w^*_{\text{org}}$$

$$\frac{2\, M_{\text{XeN}}}{M_{\text{YCore}}} = w^*_{\text{XeN}}$$

Illustrative numeric example (assumptions):

$$M_{\text{org}} = 300\ \text{g/mol}$$

$$M_{\text{XeN}} = 145\ \text{g/mol}$$

$$M_{\text{SiR2O}} = 60\ \text{g/mol}$$

$$w^*_{\text{org}} = 0.40, \quad w^*_{\text{XeN}} = 0.10, \quad w^*_{\text{Si}} = 0.50$$

Linear system to solve:

$$\frac{2 \cdot 145}{300n + 290 + 60m} = 0.10$$

$$\frac{300n}{300n + 290 + 60m} = 0.40$$

Solution (numeric):

$$n \approx 3.867$$

$$m \approx 24.17$$

Rounded conceptual baseline:

$$\boxed{n = 4, \quad m = 24}$$

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3 Effective-medium theory

3.1 Isotropic Maxwell-Garnett (multi-inclusion)

$$\boxed{\frac{\varepsilon_{\text{eff}} - \varepsilon_h}{\varepsilon_{\text{eff}} + 2\varepsilon_h} = \sum_k v_k \cdot \frac{\varepsilon_k - \varepsilon_h}{\varepsilon_k + 2\varepsilon_h}}$$

Complex permittivity (losses):

$$\tilde{\varepsilon}_i(\omega) = \varepsilon_i - j\frac{\sigma_i}{\omega\varepsilon_0}$$

$$\sigma_{\text{eff}}(\omega) = -\omega\varepsilon_0\, \Im\{\tilde{\varepsilon}_{\text{eff}}(\omega)\}$$

3.2 Bruggeman self-consistent form (three phases)

$$\boxed{\sum_k v_k \cdot \frac{\varepsilon_k - \varepsilon_{\text{eff}}}{\varepsilon_k + 2\varepsilon_{\text{eff}}} = 0}$$

3.3 Anisotropic EMT for BNNT (tensorial MG)

$$\boxed{\boldsymbol{\varepsilon}_{\text{eff}} = (\varepsilon_\parallel \quad 0 \quad 0 \backslash[4pt]0 \quad \varepsilon_\perp \quad 0 \backslash[4pt]0 \quad 0 \quad \varepsilon_\perp)}$$

$$\boxed{\alpha_\ell = \frac{\varepsilon_{\text{BNNT},\ell} - \varepsilon_h}{\varepsilon_h + L_\ell(\varepsilon_{\text{BNNT},\ell} - \varepsilon_h)}, \quad \ell \in \{\parallel, \perp\}}$$

$$\boxed{\frac{\varepsilon_{\text{eff},\ell} - \varepsilon_h}{\varepsilon_h} = \frac{v_{\text{BNNT}}\alpha_\ell}{1 - v_{\text{BNNT}}\alpha_\ell} + \sum_{k\neq\text{BNNT}} v_k \cdot \frac{\varepsilon_k - \varepsilon_h}{\varepsilon_k + 2\varepsilon_h}}$$

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4 Microstructure inductance

$$L_{\text{mat}}$$

and resonator condition

4.1 Capacitance

$$\boxed{C_{\text{mat}} = \varepsilon_0\, \varepsilon_{\text{eff}}\, \frac{A}{d}}$$

4.2 Microstructure inductance model (parametric)

$$\boxed{L_{\text{mat}} = L_{\text{self}} + \sum_{i\neq j} M_{ij} + L_{\text{kinetic}}}$$

$$\boxed{L_{\text{self}} \approx \mu_0\, \mu_r\, \frac{N^2\, A_{\text{loop}}}{l_{\text{eff}}}}$$

$$\boxed{M_{ij} \approx k_{ij}\sqrt{L_i\, L_j}, \quad 0 < k_{ij} < 1}$$

$$\boxed{L_{\text{kinetic}} \approx \frac{m^*}{n_e e^2} \cdot \frac{l}{A_c}}$$

4.3 Resonance condition

$$\boxed{f_0 = \frac{1}{2\pi\sqrt{L_{\text{mat}}\, C_{\text{mat}}}} = 13.56 \times 10^6\ \text{Hz}}$$

$$\boxed{L_{\text{mat}} = \frac{1}{(2\pi f_0)^2\, C_{\text{mat}}}}$$

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5 Losses, Q-factor, and feedback loop stability

5.1 Complex permittivity and Q

$$\boxed{\tilde{\varepsilon}_{\text{eff}}(\omega) = \varepsilon'_{\text{eff}}(\omega) - j\,\varepsilon''_{\text{eff}}(\omega)}$$

$$\boxed{\varepsilon''_{\text{eff}}(\omega) = \frac{\sigma_{\text{eff}}(\omega)}{\omega\varepsilon_0}}$$

$$\boxed{Q(\omega_0) \approx \frac{\varepsilon'_{\text{eff}}(\omega_0)}{\varepsilon''_{\text{eff}}(\omega_0)} = \frac{\omega_0\varepsilon_0\varepsilon'_{\text{eff}}(\omega_0)}{\sigma_{\text{eff}}(\omega_0)}}$$

5.2 Material transfer function

$$\boxed{H_{\text{mat}}(\omega) = \frac{K_{\text{mat}}}{1 + j\,Q^{-1}\left(\dfrac{\omega}{\omega_0} - \dfrac{\omega_0}{\omega}\right)}}$$

5.3 Feedback path and Barkhausen conditions

$$\boxed{H_{\text{fb}}(\omega) = K_{\text{fb}}(\omega)\,e^{-j\omega\tau_{\text{fb}}}}$$

$$\boxed{|G(\omega_0)| = |H_{\text{mat}}(\omega_0)|\,|H_{\text{fb}}(\omega_0)| \geq 1}$$

$$\boxed{\arg H_{\text{mat}}(\omega_0) - \omega_0\tau_{\text{fb}} = 2\pi k, \quad k \in \mathbb{Z}}$$

5.4 Phase expression

$$\boxed{\arg H_{\text{mat}}(\omega) = -\arctan\left(Q \cdot \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)\right)}$$

$$\boxed{\tau_{\text{fb}} = \frac{2\pi k}{\omega_0}}$$

$$\boxed{\tau_{\text{fb}}(k=1) = \frac{1}{f_0} \approx 73.75\ \text{ns}}$$

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6 Latency constraint and geometric limits

6.1 Latency formula

$$\boxed{t_p = \frac{L_{\text{path}}}{v_p} = \frac{L_{\text{path}}}{c/\sqrt{\varepsilon_{\text{eff}}}} \leq 0.82 \times 10^{-9}\ \text{s}}$$

6.2 Maximum path length

$$\boxed{L_{\text{path,max}} = 0.82 \times 10^{-9} \cdot \frac{c}{\sqrt{\varepsilon_{\text{eff}}}}}$$

6.3 Example

$$\boxed{\varepsilon_{\text{eff}} = 6.10 \quad \Rightarrow \quad L_{\text{path,max}} \approx 0.099\ \text{m}}$$

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7 Optimization problem, solver notes, manufacturability constraints

7.1 Mathematical program

$$\boxed{\mathbf{x} = \{v_{\mathrm{BNNT}}, v_{\mathrm{GaN}}, d, A, n, m, L_{\mathrm{mat}}, C_{\mathrm{mat}}\}}$$

$$\boxed{\min_{\mathbf{x}} \ \mathrm{Cost}(\mathbf{x}) = \alpha_{\mathrm{mat}} \sum_k v_k c_k + \alpha_{\mathrm{proc}} \cdot \mathrm{ProcCost}(d, A)}$$

subject to

$$\boxed{f_0(\mathbf{x}) = 13.56 \times 10^6, \ \backslash[4pt] \qquad t_p(\mathbf{x}) \le 0.82 \times 10^{-9}, \ \backslash[4pt] |G(\omega_0; \mathbf{x})| \ge 1, \ \backslash[4pt] \qquad v_k^{\min} \le v_k \le v_k^{\max}, \quad d_{\min} \le d \le d_{\max}.}$$

7.2 Solver notes

Use nested evaluation: outer optimizer (SLSQP/trust-region) calls inner root-finder for $\varepsilon_{\mathrm{eff}}$.

Provide analytic gradients where possible; otherwise finite differences.

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8 Worked numerical example (expanded)

Assumptions (illustrative):

$$\varepsilon_h = 6.0, \quad \varepsilon_{\mathrm{BNNT}\parallel} = 4.0, \quad \varepsilon_{\mathrm{BNNT}\perp} = 3.0, \quad \varepsilon_{\mathrm{GaN}} = 9.0$$

$$v_{\mathrm{BNNT}} = 0.10, \quad v_{\mathrm{GaN}} = 0.15, \quad v_{\mathrm{Si}} = 0.75$$

$$A = 1 \times 10^{-6}\ \mathrm{m}^2, \quad d = 1 \times 10^{-6}\ \mathrm{m}, \quad f_0 = 13.56 \times 10^6\ \mathrm{Hz}$$

Step B — isotropic MG RHS:

$$\boxed{\mathrm{RHS} = 0.10 \cdot \frac{3.5 - 6.0}{3.5 + 12.0} + 0.15 \cdot \frac{9.0 - 6.0}{9.0 + 12.0} \approx -0.016129 + 0.0214286 = 0.0052996}$$

Solve for (\varepsilon_{\text{eff}}):

$$\boxed{\frac{\varepsilon_{\mathrm{eff}} - 6.0}{\varepsilon_{\mathrm{eff}} + 12.0} = 0.0052996 \quad \Rightarrow \quad \varepsilon_{\mathrm{eff}} \approx 6.096}$$

Capacitance:

$$\boxed{C_{\mathrm{mat}} = \varepsilon_0 \varepsilon_{\mathrm{eff}} \frac{A}{d} \approx 8.854 \times 10^{-12} \cdot 6.096 \cdot \frac{1 \times 10^{-6}}{1 \times 10^{-6}} \approx 5.40 \times 10^{-11}\ \mathrm{F}}$$

Required inductance:

$$\boxed{L_{\mathrm{mat}} = \frac{1}{(2\pi f_0)^2 C_{\mathrm{mat}}} \approx 2.54 \times 10^{-6}\ \mathrm{H}}$$

Q estimate (illustrative):

$$\boxed{Q \approx \frac{\omega_0 \varepsilon_0 \varepsilon_{\mathrm{eff}}}{\sigma_{\mathrm{eff}}} = \frac{2\pi \cdot 13.56 \times 10^6 \cdot 8.854 \times 10^{-12} \cdot 6.096}{1 \times 10^{-3}} \approx 4.6 \times 10^1}$$

Feedback magnitude (illustrative):

$$\boxed{|G| \approx 1.2 \quad (\text{example: } K_{\mathrm{mat}} \approx 1, \ K_{\mathrm{fb}} = 1.2)}$$

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9 Monte-Carlo plan and convergence metrics

Parameter distributions (example):

$$\varepsilon_{\mathrm{BNNT}\parallel} \sim \mathcal{N}(4.0, 0.3^2)$$

$$\varepsilon_{\mathrm{BNNT}\perp} \sim \mathcal{N}(3.0, 0.2^2)$$

$$\varepsilon_{\mathrm{GaN}} \sim \mathcal{N}(9.0, 0.3^2)$$

$$v_{\mathrm{BNNT}} \sim \mathcal{U}(0.05, 0.15)$$

$$d \sim \mathcal{U}(0.95 d_0, 1.05 d_0)$$

Sampling plan:

$$N = 10{,}000$$

Feasibility metric:

$$\left| \frac{f(\mathbf{x}) - f_0}{f_0} \right| \leq 0.01, \quad t_p \leq 0.82 \text{ ns}, \quad |G| \geq 1$$

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10 Assumptions table, safety/regulatory notes, minimal validation checklist

10.1 Assumptions table (example entries)

Parameter	Symbol	Value (example)	Unit	Source / Status
Host permittivity	ε_h	6.0	—	assumed
BNNT permittivity (parallel)	$\varepsilon_{\mathrm{BNNT}\parallel}$	4.0	—	assumed
BNNT permittivity (perp)	$\varepsilon_{\mathrm{BNNT}\perp}$	3.0	—	assumed
GaN permittivity	$\varepsilon_{\mathrm{GaN}}$	9.0	—	literature/assumed
Effective conductivity	σ_{eff}	1×10^{-3}	S/m	assumed
Cell area	A	1×10^{-6}	m²	design choice

10.2 Safety & regulatory notes

Obtain MSDS for BNNT and exotic compounds (e.g., XeN).

Evaluate EHS risks for nanopowders and nanomaterials.

10.3 Minimal experimental validation checklist

Measure $\tilde{\varepsilon}(\omega)$ and $\sigma(\omega)$ for each phase at and around 13.56 MHz.

Validate EMT on coupons.

EM-simulate representative BNNT bundle to estimate L_{mat} and M_{ij}.

11 Short summary and next steps

Summary: This document uses `$$` for every displayed equation. It includes anisotropic EMT for BNNT, a parametric microstructure inductance derivation for

$$L_{\mathrm{mat}}$$

, explicit complex-permittivity → Q linkage, quantitative phase/delay relations, and a concrete Monte-Carlo plan with convergence metrics. An Assumptions Table and safety/regulatory checklist are included.

Next steps (choose one):

1. Produce the fully expanded Markdown file (single code block) with explicit depolarization factor formulas

   $$L_{\parallel}, L_{\perp}$$

   , a larger Assumptions Table, and Monte-Carlo pseudocode.
2. Produce the same document with the Assumptions Table filled using literature-plausible constants (marked "assumed").
3. Produce only the anisotropic EMT +

   $$L_{\mathrm{mat}}$$

   derivation sections expanded.

Confirm option 1, 2, or 3 and I will generate the requested file now.